

Devashish Dhaulakhandi

GPU Systems · Program Analysis · Compilers

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EDUCATION

Indian Institute of Science (IISc), Bengaluru

M.Tech. (Research), Computer Science and Engineering

Jul 2025 – Present

Advisor: Prof. Arkaprava Basu

VIT Bhopal University

B.Tech., Computer Science and Engineering

Aug 2021 – Jul 2025

CGPA: 9.13/10

RESEARCH EXPERIENCE

GPU Program Analysis

Research, Indian Institute of Science

Ongoing

- Developing program analysis techniques to identify correctness and performance issues in GPU programs.
- Building analyses over LLVM/NVVM IR to reason about control flow, data flow, and memory behavior in CUDA programs.

GPU Memory Safety: Trinity

Research, Indian Institute of Science

Sep 2026

- Contributed to **Trinity**, a static analysis tool for GPU race detection and memory safety; manuscript submitted for publication.

SuperCollider

CUDA, LLVM Instrumentation, GPU Concurrency

Aug 2026

- Reimplemented **SuperCollider** from the NVIDIA Research PLDI 2026 paper using the published design, without access to a public implementation.
- Built LLVM instrumentation and a GPU runtime to perturb memory operation ordering and expose races.
- Extended the implementation to cover asynchronous memory operations and additional GPU execution patterns; evaluated the tool across diverse GPU concurrency scenarios. [GitHub]

EXPERIENCE

Space Applications Centre, ISRO, Ahmedabad

Research Intern, Navigation and Receivers' Division

Nov 2024 – Mar 2025

- Implemented a Differential GNSS pipeline in C using simulator and satellite receiver-ASIC data, including pseudorange correction and least-squares position estimation.
- Achieved approximately **0.18 m 3D RMS with NavIC** and **0.30 m 3D RMS with GPS** accuracy over six-hour code-phase experiments.
- Implemented an RTCM 10403.3-based receiver-to-receiver correction protocol over UART, with serialization, buffering, asynchronous receive-side processing, and deployment on a multi-receiver setup.

Centre for Artificial Intelligence & Robotics, DRDO, Bengaluru

Project Trainee, Information and Systems Division

Jul 2024 – Sep 2024

- Analyzed network-flow data using graph-based methods and optimized queries over **2M+ rows**, reducing query execution times from approximately **40 min to 2 min**.
- Built a C++ micro-batching connector for PostgreSQL-to-Neo4j ingestion, reducing a representative 2,000+ row pipeline step from approximately **0.8 s to 0.073 s**.

SYSTEMS & COMPILER PROJECTS

Linux Virtual Memory 2026

Linux Kernel, Page Tables, Memory Management

- Extended Linux virtual memory code for inspecting and modifying user mappings through page table walks, PTE updates, reference handling, and address space synchronization.
- Implemented page deduplication support to identify identical page contents and merge mappings while preserving page references and mapping correctness. [Kernel work]

Linux Tracing with eBPF 2026

eBPF, libbpf, Kernel Instrumentation

- Built an eBPF/libbpf tracer for process lifecycle and file activity, including process creation, termination, and file operations.
- Implemented kernel-side probes and a userspace event collection pipeline for processing and reporting traced events. [GitHub]

Anticipated Expressions & Code Hoisting 2026

LLVM, Dataflow Analysis

- Implemented backward anticipated expressions analysis in LLVM using a classical per-basic-block GEN/KILL approach and fixed-point dataflow propagation.
- Used the analysis to identify safe opportunities for code hoisting, with checks for speculative execution and operand dominance before transformation. [GitHub]

Affine Loop Interchange 2026

MLIR, Affine Dialect, Loop Transformations

- Implemented an MLIR pass for affine loop interchange, including loop-nest analysis and transformations.
- Used locality and parallelism properties to choose among legal permutations for better performance. [GitHub]

CUDA Programming & GPU Performance 2025–2026

CUDA C/C++, Memory Hierarchy, Nsight

- Implemented and optimized 2D convolution kernels using coalesced memory accesses, shared-memory tiling, and data reuse.
- Used **Nsight Compute** and **Nsight Systems** to analyze memory traffic, kernel execution, and performance bottlenecks; validated GPU output against a CPU reference. [GitHub]

TECHNICAL SKILLS

Languages: C, C++, CUDA C/C++, Python, SQL

Compilers & Program Analysis: LLVM, LLVM IR, NVVM, MLIR, static analysis, dataflow analysis, control flow analysis

GPU Systems: CUDA, GPU memory hierarchy, concurrency, race detection, memory safety

Systems: Linux kernel, virtual memory, page tables, eBPF/libbpf, multithreading, synchronization

Profiling & Development: Nsight Compute, Nsight Systems, CMake, Git

Data Systems: PostgreSQL, Neo4j

ACHIEVEMENTS

- **GATE CS & IT 2024: All India Rank 175, Score 831.**